1 PRELIM WALL LAYOUT - STORY DRIFT GOVERNS

Check	Lateral system proof of concept
Limit	$\Delta < L/400$
Governing load case	$1.0W$ 25-year wind = 87 mph

2 FOUNDATION PRELIM DESIGN

Check	Foundation prelim design
Limit	$q_a = 2$ tsf, 3 tsf, etc.
Governing load case	ASD earthquake and wind combinations

ASD: EQ

$1.0D + 0.7E$
$1.0D + 0.525E + 0.75L$
$0.6D + 0.7E$

ASD: WIND

700-year wind = 36,500 weeks, 117 mph
$1.0D + 0.6W$
$1.0D + 0.75L + 0.45W$
$0.6D + 0.6W$

3 SHEAR WALL DESIGN

Check	Design steel reinforcement in shear wall
Limit	Consider cracking limit f_r
Governing load case	LRFD earthquake and wind combinations

$$f_r = 7.5\lambda\sqrt{f'_c} \tag{1}$$

f_r modulus of rupture [psi]
λ lightweight concrete modification factor, 1.0 normal weight
f'_c specified concrete compressive strength [psi]

$$A_{s,\mathrm{req}} = \frac{0.85 f'_c b}{f_y}\left[d - \sqrt{d^2 - \frac{2M_u}{0.85\,\phi\, f'_c\, b}}\right] \tag{2}$$

$A_{s,\mathrm{req}}$ required steel area [in^2]
b wall thickness or compression width [in]
f_y yield strength of longitudinal reinforcement [psi]
d effective depth to tension steel centroid [in]
M_u factored moment demand [lb-in]

LRFD: EQ

$1.2D + 1.0E$
$0.9D + 1.0E$

LRFD: WIND

$1.2D + 1.0W + 1.0L$
$0.9D + 1.0W$

4 CHECK DEFLECTION

Check	Deflection
Limit	$\Delta < L/400$
Governing load case	$1.0D + 1.0W + 0.5L$